A closed form for $[R_e(t), R_{e'}(t)]$, and the sharpness of the rigid factor

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Abstract

Let $R_e(t)$ be the operator on symmetric functions that adds an e-ribbon with weight t^{ht}. Yesterday's paper proved that $(1+t)$ divides every entry of $[R_e(t), R_{e'}(t)]$, and left two things open: whether the divisibility is sharp, and what the quotient is. Both are settled here, by one computation. Passing to the Maya diagram, where R_e becomes a weighted bead hop $b \mapsto b+e$, the commutator is a sum over *two-hop paths*, and these fall into exactly two families. The result (Theorem 10) is that $[R_e, R_{e'}]$ is supported on two kinds of pair (λ, μ), with

$$[R_e, R_{e'}]_{\mu\lambda} = (\alpha - \beta)\, t^{h-1}(1+t) \quad \text{or} \quad \pm\, t^{H-1}(1-t^2),$$

$\alpha, \beta \in \{0,1\}$, and is zero on everything else. Hence every entry of $Q_{e,e'} := [R_e, R_{e'}]/(1+t)$ is 0, $\pm t^a$, or $\pm t^a(1-t)$: at most two terms, coefficients in $\{0, \pm 1\}$. Two witness families, uniform in $e < e'$ — $s_\varnothing \mapsto s_{(e',1^e)}$ with entry $t^{e-1}(1+t)$, and $s_\varnothing \mapsto s_{(e',e)}$ with entry $1-t^2$ — show the greatest common divisor of the entries is *exactly* $t+1$ (Corollary 13), which was the open half of the programme's rigidity statement. The geometric content is a commutation criterion: two ribbon additions commute, *as graded operators*, unless the intervals of diagonals they occupy properly cross (Corollary 15). Finally, a bounded prior-art phase settles the status of yesterday's Theorem A: it is **classical**, being — up to the involution ω and a global sign, a dictionary I verified in my own engine on two independent routes — the q-Murnaghan–Nakayama rule for Iwahori–Hecke algebra characters (§7).

1 What was open

The standing result is `proofs/2026-09-03-Q75-multiplication-defect.tex`. Recall the object. For $0 \le h \le e-1$ let $N_e^{(h)}$ be the operator on the ring Λ of symmetric functions over $\mathbb{Z}[t]$ with $N_e^{(h)} s_\lambda = \sum_\mu s_\mu$, summed over those μ obtained from λ by **adding** an e-ribbon spanning exactly $h+1$ rows, and put

$$R_e(t) = \sum_{h=0}^{e-1} t^h N_e^{(h)}, \qquad \text{so} \qquad R_e(t)_{\mu\lambda} = \begin{cases} t^{h(\mu/\lambda)} & \mu/\lambda \text{ an } e\text{-ribbon}, \\ 0 & \text{else.} \end{cases}$$

Here an e-ribbon (rim hook, border strip) is a connected skew shape of size e with no 2×2 square, and its height h is one less than its number of rows.

Yesterday proved $R_e(-1) = M_{p_e}$, multiplication by the power sum, and deduced from the commutativity of Λ that $(1+t)$ divides every entry of $[R_e(t), R_{e'}(t)]$. Two questions were left open, and are the subject of this paper.

(S) *Sharpness.* Is the greatest common divisor of the entries of $[R_e, R_{e'}]$ exactly $t+1$, rather than a higher power? Equivalently, is $Q_{e,e'} := [R_e, R_{e'}]/(1+t)$ nonzero at $t = -1$? Observed on three pairs (e, e'); not proved.

(C) *Closed form.* What is $Q_{e,e'}$?

Both are answered by Theorem 10. The move that makes them finite is to stop working with skew shapes and work with beads.

2 The bead model

Definition 1. For a partition λ put $B(\lambda) = \{\lambda_i - i : i \geq 1\} \subset \mathbb{Z}$ (with $\lambda_i = 0$ for $i > \ell(\lambda)$). This is the *Maya diagram* of λ: it contains every sufficiently negative integer and omits every sufficiently large one. Writing its elements in decreasing order $b_1 > b_2 > \cdots$, one recovers $\lambda_i = b_i + i$, so $\lambda \mapsto B(\lambda)$ is injective.

The following is classical (it is the abacus of James–Kerber); I give the proof because the programme's height convention is the part of the object that is mine, and because the last fortnight's failures were all in the evidence layer. Nothing below is quoted from memory.

Proposition 2 (bead hops)**.** *Let λ be a partition, $B = B(\lambda)$, and $e \geq 1$.*

(i) The partitions μ such that μ/λ is an e-ribbon are exactly the partitions with $B(\mu) = (B \setminus \{b\}) \cup \{b + e\}$ for some $b \in B$ with $b + e \notin B$; and b is determined by μ, being the unique element of $B(\lambda) \setminus B(\mu)$.

(ii) For such a μ, the height is $h(\mu/\lambda) = \#\big(B \cap (b, b+e)\big)$, the number of beads strictly jumped over.

(iii) The cells of μ/λ have contents (i.e. values of $j - i$ on a cell in row i, column j) exactly $b+1, b+2, \ldots, b+e$, each once.

(iv) Listing the cells of μ/λ as $c_1, \ldots, c_e$ in increasing content — which is the order along the ribbon from its bottom-left end — the cell c_{j+1} lies directly above c_j if $b + j \in B$, and directly to the right of c_j if $b + j \notin B$.

Proof. Let $b \in B$ with $b + e \notin B$, let $k = \#\big(B \cap (b, b+e)\big)$, and let $b = b_r$ in the decreasing enumeration of B. The elements of B lying in $(b, b+e)$ are then exactly $b_{r-1}, b_{r-2}, \ldots, b_{r-k}$, and $b_{r-k-1} > b + e$ (using $b + e \notin B$). Set $B' = (B \setminus \{b\}) \cup \{b + e\}$. Its decreasing enumeration is

$$b_1,\ \ldots,\ b_{r-k-1},\quad \underbrace{b+e}_{\text{position } r-k},\ b_{r-k},\ b_{r-k+1},\ \ldots,\ b_{r-1},\ b_{r+1},\ \ldots,$$

so, writing μ for the partition with $B(\mu) = B'$ and using $\mu_i = b'_i + i$,

$$\mu_i = \lambda_i\ (i < r-k \text{ or } i > r), \qquad \mu_{r-k} = b + e + r - k, \qquad \mu_i = \lambda_{i-1} + 1\ (r - k < i \leq r). \tag{1}$$

In particular $\mu \supseteq \lambda$: for $r - k < i \leq r$ we have $\mu_i = \lambda_{i-1} + 1 > \lambda_i$, and $\mu_{r-k} = b + e + r - k > b_{r-k} + r - k = \lambda_{r-k}$ because $b_{r-k} < b + e$.

Now read off μ/λ from (1). It is empty outside rows $r-k,\dots,r$. In row i with $r-k<i\le r$ it occupies the columns $\lambda_i+1,\dots,\lambda_{i-1}+1$, which is $\lambda_{i-1}+1-\lambda_i=b_{i-1}-b_i\ge 1$ cells; in row $r-k$ it occupies the columns $\lambda_{r-k}+1,\dots,b+e+r-k$, which is $b+e-b_{r-k}\ge 1$ cells. Summing,

$$|\mu/\lambda| = (b+e-b_{r-k}) + \sum_{i=r-k+1}^{r}(b_{i-1}-b_i) = (b+e-b_{r-k})+(b_{r-k}-b_r)=e,$$

since $b_r=b$. (When $k=0$ the middle sum is empty and the single row r has $b+e-b_r=e$ cells.) Rows $i-1$ and i, for $r-k<i\le r$, share exactly the column $\lambda_{i-1}+1$ — it is the last column of row i and the first of row $i-1$ — and rows i and $i-2$ share none, since row i stops at column $\lambda_{i-1}+1$ and row $i-2$ starts at column $\lambda_{i-2}+1>\lambda_{i-1}+1$. A skew shape whose nonempty rows are consecutive and whose consecutive rows overlap in exactly one column is connected and contains no 2×2 square; so μ/λ is an e-ribbon, and it occupies $k+1$ rows, giving (ii).

For (iii): the contents occupied in row i ($r-k<i\le r$) are $(\lambda_i+1)-i,\dots,(\lambda_{i-1}+1)-i$, i.e. the integers in $(b_i,b_{i-1}]$; and in row $r-k$ they are the integers in $(b_{r-k},b+e]$. These intervals are consecutive and disjoint, and their union is $(b_r,b+e]=(b,b+e]$. This proves (iii), and shows that increasing content is the same as travelling along the ribbon from row r to row $r-k$, i.e. from the bottom-left end. For (iv): within a row the content increases by 1 as the column does, so c_{j+1} is to the right of c_j unless c_j is the last cell of its row. By the description just given, the last cell of row i has content b_{i-1}, and the first cell of row $i-1$ — which is the next one in content order — lies in column $\lambda_{i-1}+1$, the same column. So the ribbon steps up between contents d and $d+1$ exactly when $d\in\{b_{r-1},\dots,b_{r-k}\}=B\cap(b,b+e)$. Since c_j has content $b+j$, this is the statement of (iv).

Finally, the converse half of (i). Let μ/λ be an e-ribbon, occupying rows $r-k,\dots,r$ for some $k\ge 0$. Consecutive rows of a ribbon overlap in exactly one column, which forces $\mu_i=\lambda_{i-1}+1$ for $r-k<i\le r$, and $\mu_i=\lambda_i$ off $[r-k,r]$. Hence $b'_i=b_{i-1}$ for $r-k<i\le r$ and $b'_i=b_i$ for $i<r-k$ and $i>r$, so $B(\mu)$ and $B(\lambda)$ differ in exactly one element each: $B(\lambda)\setminus B(\mu)=\{b_r\}$ and $B(\mu)\setminus B(\lambda)=\{b'_{r-k}\}$. Comparing sizes, $|\mu|-|\lambda|=b'_{r-k}-b_r=e$. Setting $b=b_r$ gives $B(\mu)=(B\setminus\{b\})\cup\{b+e\}$, and $b+e\notin B$ because $b+e=b'_{r-k}\notin B(\lambda)$. $\square$

Corollary 3. *Identify s_λ with $B(\lambda)$. Then*

$$R_e(t)\,|B\rangle = \sum_{\substack{b\in B\\ b+e\notin B}} t^{\#(B\cap(b,b+e))}\,\big|(B\setminus\{b\})\cup\{b+e\}\big\rangle.$$

Everything from here on is a statement about beads hopping to the right on $\mathbb{Z}$.

3 Two-hop paths

Definition 4. A *hop* is a pair (u,d) with $u\in\mathbb{Z}$ and $d\ge 1$. It is *applicable* to a Maya diagram B if $u\in B$ and $u+d\notin B$, and it then produces $B\cdot(u,d)=(B\setminus\{u\})\cup\{u+d\}$ with *weight* $t^{\#(B\cap(u,u+d))}$.

Fix $e\ne e'$ and Maya diagrams B,B'. A *path* from B to B' is a sequence of two hops $(u_1,d_1),(u_2,d_2)$ with $(d_1,d_2)\in\{(e',e),(e,e')\}$, the first applicable to B, the second to $B_1:=B\cdot(u_1,d_1)$, and $B_1\cdot(u_2,d_2)=B'$. Its *sign* is $+1$ if $(d_1,d_2)=(e',e)$ and -1 if $(d_1,d_2)=(e,e')$, and its weight is the product of the two hop weights.

By Corollary 3, expanding $R_eR_{e'}-R_{e'}R_e$ (the rightmost operator acts first) gives at once:

Lemma 5. $[R_e,R_{e'}]_{\mu\lambda}=\sum_P \operatorname{sign}(P)\cdot\operatorname{wt}(P)$, *over all paths P from $B(\lambda)$ to $B(\mu)$.*

Lemma 6 (trichotomy)**.** *Let $P = \big((u_1, d_1), (u_2, d_2)\big)$ be a path from B to B', and put $v_i = u_i + d_i$. Exactly one of the following holds.*

(F) $u_2 = v_1$ *(the same bead hops twice). Then $B \setminus B' = \{u_1\}$ and $B' \setminus B = \{u_1 + e + e'\}$.*

(B) $v_2 = u_1$ *(the second bead lands on the site the first vacated). Then $B \setminus B' = \{u_2\}$ and $B' \setminus B = \{u_2 + e + e'\}$.*

(S) *neither. Then $u_1 \neq u_2$, $v_1 \neq v_2$, $B \setminus B' = \{u_1, u_2\}$ and $B' \setminus B = \{v_1, v_2\}$.*

Proof. $B_1 = (B \setminus \{u_1\}) \cup \{v_1\}$ and $B' = (B_1 \setminus \{u_2\}) \cup \{v_2\}$. Applicability of the second hop says $u_2 \in B_1$ and $v_2 \notin B_1$. If $u_2 = v_1$ we are in (F), and then $B' = (B \setminus \{u_1\}) \cup \{v_2\}$ with $v_2 = u_1 + d_1 + d_2 = u_1 + e + e'$; note $v_2 \neq u_1$ since $e + e' > 0$. Otherwise $u_2 \in B$ and $u_2 \neq u_1$. If moreover $v_2 = u_1$ we are in (B), and $B' = (B \setminus \{u_2\}) \cup \{v_1\}$ with $v_1 = u_1 + d_1 = v_2 + d_1 = u_2 + d_2 + d_1 = u_2 + e + e'$. Otherwise $v_2 \notin B$ and $v_2 \neq v_1$, giving (S). Cases (F) and (B) are exclusive: they would give $v_2 = u_1 < v_1 = u_2 < v_2$. ∎

So the support of $[R_e, R_{e'}]$ is confined to pairs (λ, μ) with $\#\big(B(\lambda) \setminus B(\mu)\big) \in \{1, 2\}$, and the two values are handled separately.

The ribbon case

Lemma 7. *Suppose $B \setminus B' = \{b\}$. Then $B' \setminus B = \{b + e + e'\}$ (else there is no path), and, writing $N = \#\big(B \cap (b, b + e + e')\big)$, $\alpha = \mathbf{1}[b + e \in B]$, $\beta = \mathbf{1}[b + e' \in B]$,*

$$\sum_P \operatorname{sign}(P) \operatorname{wt}(P) = (\alpha - \beta)\, t^{N-1}(1 + t).$$

Moreover $N \geq 1$ whenever $\alpha \neq \beta$, so the exponent is non-negative.

Proof. By Lemma 6 every path is of type (F) or (B), and in both cases the vacated site is b and the filled site is $b+e+e'$. There are four candidates, indexed by the type and by $(d_1, d_2) \in \{(e', e), (e, e')\}$.

Type (F), (d_1, d_2). The hops are (b, d_1) then $(b + d_1, d_2)$. Applicability: $b \in B$ holds; $b + d_1 \notin B$ is a genuine condition; then $b + d_1 \in B_1$ holds and $b + e + e' \notin B_1$ holds because $b + e + e' \notin B$ and $b + e + e' \neq b + d_1$. So this path exists iff $b + d_1 \notin B$. Its weight is $t^{\#(B \cap (b, b+d_1))} \cdot t^{\#(B_1 \cap (b+d_1, b+e+e'))}$, and $B_1 = (B \setminus \{b\}) \cup \{b + d_1\}$ agrees with B on the open interval $(b + d_1, b + e + e')$; hence the weight is $t^{\#(B \cap (b,b+d_1)) + \#(B \cap (b+d_1, b+e+e'))} = t^{N - \mathbf{1}[b+d_1 \in B]} = t^N$.

Type (B), (d_1, d_2). Here $u_2 = b$, $v_2 = b + d_2$, $u_1 = v_2 = b + d_2$ and $v_1 = b + e + e'$. The hops are $(b + d_2, d_1)$ then (b, d_2). Applicability: $b + d_2 \in B$ is a genuine condition; $b + e + e' \notin B$ holds; then $b \in B_1$ holds and $b + d_2 \notin B_1$ holds since $b + d_2$ was removed and $b + d_2 \neq b + e + e'$. So this path exists iff $b + d_2 \in B$. Its weight is $t^{\#(B \cap (b+d_2, b+e+e'))} \cdot t^{\#(B_1 \cap (b, b+d_2))}$, and B_1 agrees with B on $(b, b + d_2)$; hence the weight is $t^{N - \mathbf{1}[b+d_2 \in B]} = t^{N-1}$.

Collecting, with sign $+$ for $(d_1, d_2) = (e', e)$ and $-$ for (e, e'):

$$\big((1 - \beta)t^N + \alpha t^{N-1}\big) - \big((1 - \alpha)t^N + \beta t^{N-1}\big) = (\alpha - \beta)\big(t^N + t^{N-1}\big) = (\alpha - \beta)t^{N-1}(1 + t).$$

If $\alpha \neq \beta$ then one of $b + e, b + e'$ lies in B, and both lie strictly inside $(b, b + e + e')$ because $e, e' \geq 1$; so $N \geq 1$. ∎

The two-bead case

Lemma 8 (the assignment is forced)**.** *Let $e \neq e'$ and suppose $B \setminus B' = \{p, q\}$ and $B' \setminus B = \{x, y\}$ with $p \neq q$, $x \neq y$. There is at most one bijection $\phi : \{p, q\} \to \{x, y\}$ whose displacement multiset $\{\phi(p) - p,\ \phi(q) - q\}$ equals $\{e, e'\}$.*

Proof. Suppose both bijections work: $\{x - p,\ y - q\} = \{e, e'\}$ and $\{y - p,\ x - q\} = \{e, e'\}$. If $x - p = e$ then $y - p \neq e$ (else $x = y$), so $y - p = e'$; since $e \neq e'$ the second condition forces $x - q = e$, whence $p = q$. The case $x - p = e'$ is symmetric. $\square$

Lemma 9. *Let $e \neq e'$, suppose $B \backslash B'$ has two elements, and suppose the bijection of Lemma 8 exists; write b for the bead displaced by e and c for the bead displaced by e', so $B' = (B \backslash \{b, c\}) \cup \{b+e, c+e'\}$. Put $H = \#\big(B \cap (b, b+e)\big) + \#\big(B \cap (c, c+e')\big)$. Then*

$$\sum_P \operatorname{sign}(P) \operatorname{wt}(P) = \begin{cases} +t^{H-1}(1-t^2) & \text{if } b < c < b+e < c+e', \\ -t^{H-1}(1-t^2) & \text{if } c < b < c+e' < b+e, \\ 0 & \text{otherwise,} \end{cases}$$

and $H \geq 1$ in the first two cases. If the bijection does not exist, the sum is 0.

Proof. By Lemmas 6 and 8 there are at most two paths, differing only in the order of the two hops (b, e) and (c, e'). Both exist. Indeed $b, c \in B$ and $b+e, c+e' \notin B$ (they lie in $B' \backslash B$), and $b+e \neq c+e'$; for the order "c first" one needs $c + e' \notin B$ and then $b + e \notin B_c$ where $B_c = (B \setminus \{c\}) \cup \{c + e'\}$, which holds since $b + e \notin B$ and $b + e \neq c + e'$; the other order is symmetric. The order "c first, then b" has $(d_1, d_2) = (e', e)$ and so sign $+$.

Weights. With B_c as above and $B_b = (B \setminus \{b\}) \cup \{b + e\}$,

$$\#\big(B_c \cap (b, b+e)\big) = \#\big(B \cap (b, b+e)\big) - \delta_1 + \delta_2, \qquad \#\big(B_b \cap (c, c+e')\big) = \#\big(B \cap (c, c+e')\big) - \delta_3 + \delta_4,$$

where $\delta_1 = \mathbf{1}[c \in (b, b+e)]$, $\delta_2 = \mathbf{1}[c+e' \in (b, b+e)]$, $\delta_3 = \mathbf{1}[b \in (c, c+e')]$ and $\delta_4 = \mathbf{1}[b+e \in (c, c+e')]$. Hence the total is

$$t^H\big(t^{\delta_2 - \delta_1} - t^{\delta_4 - \delta_3}\big). \tag{2}$$

Note $b \neq c$, and $b + e \neq c$ and $c + e' \neq b$ because $B \setminus B'$ and $B' \setminus B$ are disjoint, and $b + e \neq c + e'$.

Assume first $b < c$; then $\delta_3 = 0$, $\delta_1 = \mathbf{1}[c < b + e]$, $\delta_2 = \mathbf{1}[c + e' < b + e]$ (as $c + e' > c > b$) and $\delta_4 = \mathbf{1}[c < b + e < c + e']$.

- If $c > b + e$: then $\delta_1 = 0$; also $c + e' > c > b + e$ so $\delta_2 = 0$, and $\delta_4 = 0$. By (2) the total is $t^H(1 - 1) = 0$.
- If $c < b + e$: then $\delta_1 = 1$ and $\delta_4 = \mathbf{1}[b + e < c + e']$. Since $b + e \neq c + e'$, either $c + e' < b + e$, giving $\delta_2 = 1, \delta_4 = 0$ and total $t^H(t^0 - t^0) = 0$; or $c + e' > b + e$, giving $\delta_2 = 0, \delta_4 = 1$ and total $t^H(t^{-1} - t) = t^{H-1}(1 - t^2)$. The second alternative is precisely $b < c < b + e < c + e'$, and there $c \in B \cap (b, b + e)$, so $H \geq 1$.

The case $c < b$ is obtained by exchanging the roles of (b, e) and (c, e'), which exchanges (δ_1, δ_2) with (δ_3, δ_4) and therefore negates (2). $\square$

4 The closed form

Theorem 10. *Let $e \neq e'$ and let λ, μ be partitions; write $B = B(\lambda)$, $B' = B(\mu)$. Then $[R_e(t), R_{e'}(t)]_{\mu\lambda}$ is:*

(i) *if $B \setminus B' = \{b\}$ and $B' \setminus B = \{b+e+e'\}$ — equivalently, if μ/λ is an $(e+e')$-ribbon, in which case $h := h(\mu/\lambda) = \#(B \cap (b, b+e+e'))$ —*

$$\big(\mathbf{1}[b+e \in B] - \mathbf{1}[b+e' \in B]\big)\; t^{\,h-1}\,(1+t);$$

(ii) *if $B \setminus B' = \{b, c\}$ and $B' \setminus B = \{b+e, c+e'\}$ (an assignment that is unique if it exists), with $H = \#(B \cap (b, b+e)) + \#(B \cap (c, c+e'))$ —*

$$\begin{cases} +\,t^{\,H-1}(1-t^2) & \text{if } b < c < b+e < c+e', \\ -\,t^{\,H-1}(1-t^2) & \text{if } c < b < c+e' < b+e, \\ 0 & \text{otherwise;} \end{cases}$$

(iii) 0 *in every other case.*

All exponents occurring with a nonzero coefficient are non-negative.

Proof. Lemma 5 reduces the entry to a signed sum over paths; Lemma 6 splits the paths into the case $\#(B \setminus B') = 1$, handled by Lemma 7, and the case $\#(B \setminus B') = 2$, handled by Lemmas 8 and 9; and no other value of $\#(B \setminus B')$ admits a path. In case (i) the description of μ/λ as an $(e+e')$-ribbon, and the identification of h, are Proposition 2(i)–(ii) applied with ribbon size $e+e'$. □

Corollary 11 (the shape of the quotient)**.** *Every entry of $Q_{e,e'} := [R_e, R_{e'}]/(1+t)$ is 0, or $\pm t^a$, or $\pm t^a(1-t)$ for some $a \geq 0$. In particular every entry has at most two terms and all its coefficients lie in $\{0, \pm 1\}$; the monomial entries are exactly those on $(e+e')$-ribbons.*

Proof. Divide the two cases of Theorem 10 by $1+t$, using $1-t^2 = (1-t)(1+t)$. □

This also re-proves, without reference to Λ being commutative, that $(1+t)$ divides $[R_e, R_{e'}]$: both surviving shapes carry the factor visibly.

5 Sharpness

Lemma 12 (two witness families)**.** *Let $2 \leq e < e'$. Then*

$$[R_e, R_{e'}]_{(e',1^e),\varnothing} = t^{\,e-1}(1+t), \qquad [R_e, R_{e'}]_{(e',e),\varnothing} = 1-t^2.$$

Proof. $B(\varnothing) = \mathbb{Z}_{<0}$, so every negative integer is a bead and no non-negative integer is.

First family. Take $b = -1-e$ and hop it to $b+e+e' = e'-1$. The beads strictly between are $-e, \ldots, -1$, so $h = e$. Here $b+e = -1 \in B$ and $b+e' = e'-e-1 \geq 0$ is not in B, so $\mathbf{1}[b+e \in B] - \mathbf{1}[b+e' \in B] = 1$ and Theorem 10(i) gives $t^{e-1}(1+t)$. The target: $B' = \big(\mathbb{Z}_{<0} \setminus \{-1-e\}\big) \cup \{e'-1\}$ has decreasing enumeration $e'-1, -1, -2, \ldots, -e, -e-2, -e-3, \ldots$, so by $\mu_i = b'_i + i$ we get $\mu_1 = e'$, $\mu_2 = \cdots = \mu_{e+1} = 1$ and $\mu_i = 0$ beyond; that is $\mu = (e', 1^e)$.

Second family. Take $b = -2$ hopping by e and $c = -1$ hopping by e'. Both are beads; $b+e = e-2 \geq 0$ and $c+e' = e'-1 \geq 0$ are not, and they are distinct. The interval $(b, b+e) = (-2, e-2)$

meets B only in $\{-1\}$ (as $e \geq 2$), and $(c, c+e') = (-1, e'-1)$ meets B not at all; so $H = 1$. The crossing condition $b < c < b+e < c+e'$ reads $-2 < -1 < e-2 < e'-1$, which holds because $e \geq 2$ and $e < e'$. Theorem 10(ii) gives $t^0(1-t^2) = 1-t^2$. The target: $B' = \big(\mathbb{Z}_{<0} \setminus \{-2,-1\}\big) \cup \{e-2, e'-1\}$ enumerates as $e'-1, e-2, -3, -4, \ldots$, giving $\mu_1 = e'$, $\mu_2 = e$, $\mu_i = 0$ beyond; that is $\mu = (e', e)$. $\square$

Corollary 13 (sharpness; question (S))**.** *For all $2 \leq e < e'$ the greatest common divisor of the entries of $[R_e(t), R_{e'}(t)]$ is exactly $t+1$. Equivalently $Q_{e,e'}(-1) \neq 0$; indeed $Q_{e,e'}(-1)_{(e',1^e),\varnothing} = (-1)^{e-1}$.*

Proof. Divisibility by $t+1$ is Corollary 11 (or Cor. T3 of the previous paper), so the gcd is $(t+1)g$ where g is the gcd of the entries of $Q_{e,e'}$. By Lemma 12, $Q_{e,e'}$ has an entry t^{e-1} and an entry $1-t$. Hence g divides t^{e-1}, so g is a power of t up to a unit; and g divides $1-t$, which is not divisible by t. So g is a unit and the gcd is $t+1$. Evaluating t^{e-1} at $t=-1$ gives $(-1)^{e-1} \neq 0$. $\square$

Both families are uniform in (e, e'), which was the form the brief asked for; the weaker fallback (a non-uniform witness, or only $e' = e+1$) is not needed.

6 Reading the answer on Young diagrams

Theorem 10 is stated on beads because that is where it is finite. Proposition 2(iii)–(iv) translates it back.

Corollary 14 (the ribbon entries)**.** *Let μ/λ be an $(e+e')$-ribbon of height h, and list its cells $c_1, \ldots, c_{e+e'}$ from its bottom-left end (equivalently, in increasing content). For $1 \leq j < e+e'$ set $\epsilon_j = 1$ if c_{j+1} lies directly above c_j and $\epsilon_j = 0$ if it lies directly to its right. Then*

$$[R_e, R_{e'}]_{\mu\lambda} = (\epsilon_e - \epsilon_{e'})\, t^{\,h-1}(1+t),$$

which is nonzero exactly when the ribbon turns upward at exactly one of its two cut points.

Proof. Immediate from Theorem 10(i) and Proposition 2(iv), which says $\epsilon_j = \mathbf{1}[b+j \in B]$. $\square$

The first witness family is the smallest instance: $\mu/\varnothing = (e', 1^e)$ is a hook, its cells read from the foot of the leg, and it turns upward at cut point e (the leg-to-arm corner) and rightward at cut point e' (inside the arm, since $e' > e$).

Corollary 15 (a commutation criterion)**.** *By Proposition 2(iii), the e-ribbon added by the hop $b \mapsto b+e$ occupies exactly the diagonals (contents) in the interval $(b, b+e]$. Case (ii) of Theorem 10 therefore says: two ribbon additions, of sizes e and e' and by distinct beads, contribute equally to the two orders — that is, they commute* as graded operators — *unless the two intervals of diagonals they occupy* properly cross*, meaning they overlap and neither contains the other. Disjoint intervals commute, and so do nested ones.*

The nested case is the one worth noticing, because it is the case a parity or degree argument would be least likely to predict: a small ribbon sitting entirely inside the diagonal span of a large one still commutes with it. Specialised to $e = e' = n$, the criterion reads $|b-c| \geq n$, which is the criterion recorded in the accepted answer to MathOverflow 292312 (Alexandersson's question, answered by Ziqing Xiang) for when two rim-hook removals may be swapped. I have *not* read that thread at source this session — it is recorded in my browse log of 2026-09-04 at the level of an agent's check — so I record the agreement as a consistency observation, not as a citation.

Remark 16. Two structural facts fall out. First, $[R_e, R_{e'}]$ vanishes at $t = 1$ on every two-bead entry (where the factor is $1 - t^2$) and equals ± 2 on the ribbon entries; so at $t = 1$ the commutator is supported on $(e + e')$-ribbons alone. Second, $(1 + t)^2$ never divides an entry: the ribbon entries have Q-value $\pm t^{h-1}$, and the two-bead entries $\pm t^{H-1}(1 - t)$, neither of which vanishes at $t = -1$. That is a stronger statement than Corollary 13: not merely some entry, but *every* nonzero entry of $[R_e, R_{e'}]$ is divisible by $1 + t$ exactly once.

7 Prior art

Theorem A of the previous paper is classical

The 2026-09-03 paper proved (its Theorem A) that for a skew shape S of size e, $\langle s_S, f_e(t)\rangle = t^{h(S)}(1 + t)^{c(S)-1}$ if S has no 2×2 square and 0 otherwise, where $f_e(t) = \sum_k t^k s_{(e-k,1^k)}$, and left its prior-art status "unsettled, flagged". A bounded search this session settles it: **it is classical**.

Jing and Liu, *Murnaghan–Nakayama rule...*, arXiv `2310.15730`, define in §4.2

$$(a-1)\sum_{r\geq 0} \tilde{h}_r(a)z^r = \prod_i \frac{1 - x_i z}{1 - a x_i z}, \qquad wt(\theta; t) = (t-1)^{m-1}\prod_{i=1}^{m}(-1)^{r(\xi_i)-1}t^{c(\xi_i)-1}$$

(θ a generalised border strip with components $\xi_1, \ldots, \xi_m$, of $r(\xi_i)$ rows and $c(\xi_i)$ columns), and state, as a corollary, $\tilde{h}_r(q)s_\mu = \sum_\lambda wt(\lambda/\mu; q)s_\lambda$ summed over λ with λ/μ an r-generalised border strip. I read all three displays myself in the `2310.15730` e-print source (`mn.tex`, lines 626, 728, 744), in context, not from a quotation.

Their attributions, stated precisely because the difference matters. The corollary is introduced as "The following Murnaghan–Nakayama rule (see [Ram], [Hal], [JL2]) for $H_n(q)$" — i.e. credited to Ram, *Invent. Math.* **106** (1991), Halverson, *JCTA* **71** (1995), and an earlier Jing–Liu paper. The weight lemma `t:wt1` is introduced as "similar to [HR, (6.7)]" — *similar to*, not *due to* — referring to Halverson–Ram, *Adv. Math.* **116** (1995), and Jing and Liu "furnish a proof for completeness". So the claim "classical" rests on the Ram/Halverson attribution of the *rule*, which I have not followed to its source.

One thing read at source that bears on the previous paper's other gap: Jing and Liu describe Ram's Frobenius formula as expressing Hecke-algebra characters "in terms of one-row Hall–Littlewood functions and Schur symmetric functions". That is independent corroboration, from a document I have read, that $\tilde{h}_r$ — and hence $f_e(t)$ — is a one-row Hall–Littlewood function.

I checked the dictionary myself, in my own engine, by two routes that share no code, rather than accepting the algebra second-hand:

- *Generating functions.* Their right-hand side at $a = -t$ is $E(-z)H(-tz)$; mine (Lemma gf of the previous paper) is $\sum_e(1 + t)f_e(t)z^e = H(z)E(tz)$, and applying ω and $z \mapsto -z$ carries one to the other. Coefficient by coefficient this predicts $\tilde{h}_e(-t) = (-1)^{e-1}\omega f_e(t)$, confirmed in the Schur basis for $e \leq 5$.
- *Weights.* Using $r(\xi) + c(\xi) = |\xi| + 1$, their $wt(\theta; -t)$ should equal $(-1)^{|\theta|-1}$ times my weight for the *conjugate* shape θ'. Confirmed on all 541 skew shapes μ/λ with $|\lambda| \leq 5$ and $|\mu/\lambda| \leq 4$ (527 of them with nonzero weight); the negative control that drops the global sign $(-1)^{|\theta|-1}$ disagrees on 327 of the 541.

So Theorem A is the q-Murnaghan–Nakayama rule for Iwahori–Hecke algebra characters, transported by ω. The honest caveat: **Ram (1991), Halverson (1995) and Halverson–Ram**

(1995) were not opened. None is on arXiv. The earliest source I can name from a document I have actually read is Jing–Liu §4.2 (2023), which proves the rule and points at three earlier ones. Before the word “classical” is used in anything outgoing, at least one of those three should be read at source.

Two nearby results were checked and are *not* it, which is worth recording because both were on my candidate list: Okada, arXiv `1904.03386` Cor. 6.8 (attributed there to Morris, JLMS **39** (1964) 481–488) and Fayers, arXiv `2003.07713` Prop. 3.6 both live in the shifted / Schur P–Q world and carry the weight $2^{\#\text{components}}$ with no t; they are the $t = -1$ shadow of the rule, in a different arena. Fayers’ remark on MathOverflow 337891 that he holds an unpublished Hall–Littlewood version by duality concerns that shifted rule, not this one.

Theorem 10 of this paper

Not searched. The brief capped the prior-art phase at Theorem A, and I spent the cap there. What the browse log of 2026-09-04 records is a *mechanism* for why the commutator across two different ribbon sizes is unlikely to have been asked: in the Fock-space literature the ribbon size n is fixed by the choice of $U_q(\widehat{\mathfrak{sl}}_n)$, and the Heisenberg index counts how many ribbons rather than what size, so R_e and $R_{e'}$ for $e \neq e'$ act on different Fock spaces there. That is an argument, not a search. The status of Theorem 10 is **unsearched**, and it should not be described as new.

8 Computational verification

All code is in `probes/2026-09-04-Q76/`. The engine `abacus.py` implements Corollary 3 on beta-numbers; it is checked against `symfunc.py` from `probes/2026-09-03-Q75/`, which builds R_e by direct enumeration of skew shapes on Young diagrams and shares no code with it. The commutator is then formed by composing operators, with no reference to Theorem 10.

Claim	Range	Result
Prop. 2(i)–(ii): beads vs. skew shapes, entrywise	$\lvert\lambda\rvert \leq 8$, $e = 2..6$	335/335 configs
Theorem 10, entrywise vs. brute force	$\lvert\lambda\rvert \leq 8$, all $2 \leq e < e' \leq 7$	238594/238594
of which nonzero	same	11360
two-bead (crossing) entries, shape $\pm t^a(1-t^2)$	same	8408
ribbon entries, shape $\pm t^a(1+t)$	same	2952
Prop. 2(iv): the cut reading, on Young diagrams	$\lvert\lambda\rvert \leq 6$, $2 \leq e < e' \leq 5$	2552/2552
Prop. 2(iii): the diagonal interval	$\lvert\lambda\rvert \leq 7$, $e = 2..6$	1058/1058
Lemma 12, both families	all $2 \leq e < e' \leq 8$	21/21, 21/21
§7 dictionary, generating functions	$e \leq 5$	exact
§7 dictionary, weights	541 skew shapes	541/541

The counts 8408 and 2952 agree with the two entry-shape counts computed on 2026-09-04 by an entirely different route — direct border-strip enumeration on Young diagrams, over the same range — and recorded in the browse log before this session began. That is a cross-check on the support and the shapes, not on the formula.

Negative controls, each with its variation named before it was run.

- *C1, the trivial kernel.* $e = e'$ must give the zero matrix: 120/120 configurations ($\lvert\lambda\rvert \leq 6$, $e \leq 5$). Theorem 10 does not apply there, and Lemma 8 is exactly where $e \neq e'$ enters.

- *C2, antisymmetry.* Exchanging e and e' negates every entry: 1208/1208.
- *C3, the height convention.* This is the control that tests the part of the object that is mine. Replacing h by the number of rows (i.e. $h+1$) in the bead operator must *break* Theorem 10: it disagrees on 761 of 6665 entries. The check is therefore not blind to the grading. No convention was chosen during this session; the height is the one fixed by Definition 1 and the previous paper.
- *C4, the crossing condition is load-bearing.* A variant prediction that keeps everything else but drops the crossing hypothesis in case (ii) — i.e. returns $\pm t^{H-1}(1-t^2)$ for every two-bead pair — disagrees with the truth on 12088 entries, exactly the number of reachable non-crossing pairs. So the vanishing in case (ii) is a real constraint and not an artefact of an empty support.
- *C5, the cut condition is load-bearing.* A variant that returns $t^{h-1}(1+t)$ on every $(e+e')$-ribbon, ignoring $\epsilon_e - \epsilon_{e'}$, disagrees on 7040 entries.
- *C6, the prior-art dictionary.* Dropping the global sign $(-1)^{|\theta|-1}$ disagrees on 327/541 shapes.

Every control fires. The one I would treat as decisive is C3: the two shapes $\pm t^a(1+t)$ and $\pm t^a(1-t^2)$ are statements about the height grading, and a check that survived a perturbed height would be measuring only the support.

9 Gaps

1. The prior-art status of Theorem 10 is **unsearched** (§7).
2. The identification of the previous paper's Theorem A as classical rests on Jing–Liu `2310.15730` §4.2, read at source by me, plus a dictionary I verified computationally. The onward attribution to Ram (1991), Halverson (1995) and Halverson–Ram (1995) is a citation-chain claim read inside Jing–Liu; none of those three has been opened, and none is on arXiv.
3. The Hall–Littlewood identification (`cor:HL`) of the previous paper (the identification $f_e(t) = P_{(e)}(x;-t)$) still rests on two facts quoted from Macdonald III from memory. The prior-art phase did not reach them; the partial findings are that Jing–Liu §5.2 gives the exponential form $\sum_n q_n(X;t)z^n = \exp\left(\sum_n \frac{1-t^n}{n} p_n z^n\right)$ and the identification $q_n = Q_{(n)}$, but not the product form nor $q_r = (1-t)P_{(r)}$; and that their §4.2 prose describes $\tilde{h}_r$ as a one-row Hall–Littlewood function, which corroborates the identification without supplying the two Macdonald displays.
4. Theorem 10 says nothing about the q-deformed rigid factor $(1+qt)$ of Q59, nor about the Heisenberg scalar $[e]_{t^2}$ of $[R_e^{\perp}, R_e]$. Those brackets do not vanish at $t=-1$; this one does. They are different phenomena and this paper does not connect them.
5. The two-bead entries are described by a condition on beads. Corollary 15 translates it to diagonals, but I have not characterised, intrinsically in terms of the skew shape μ/λ alone, which shapes of size $e+e'$ arise in case (ii) and with which multiplicities. For the ribbon case this is done (Corollary 14).